

DSCI 550 Group 8 Homework 3 Report

I. Justification for Selecting the Five D3 Visualizations

Q: Why did you select your 5 D3 visualizations? How are they answering and showing off your features from assignments 1 and 2 and the work you did?

A:

1. Frequency of Hauntings Across Dates
 - a. We selected the D3 gallery calendar heatmap visualization to visualize the frequency of hauntings across dates because it is the most intuitive and powerful way to capture temporal patterns at a daily resolution. This visualization allows viewers to quickly identify spikes in activity—such as those that occur around Halloween or during specific years—with color intensity directly tied to the number of hauntings reported on each day. The calendar format also offers a natural structure for identifying seasonal trends, anomalies, and gaps in data coverage. Unlike line graphs or bar charts, which require aggregation over larger intervals, the calendar heatmap preserves granularity while still offering a high-level overview through its color gradients. Through analyzing this visualization, we discovered that the month with the highest average occurrences is January, the year with the most total occurrences was 2025 with 1652 occurrences in total, and the maximum occurrences on a single day was 1230. This visualization builds directly on our work in Assignments 1 and 2, where we cleaned and standardized the “Haunted Places Date” field and grouped occurrences by date. By resolving inconsistencies through datetime conversion and summarizing hauntings temporally, we transformed a noisy column into a meaningful visual. The D3 gallery calendar highlights real trends in reported hauntings, demonstrating the impact of our data wrangling and feature extraction.
2. Most Common Apparition Types Over the Year
 - a. We chose this visualization to see if there were any trends in the most common types of apparitions across all the years from 1900-2024. The four main types we documented were: ghost, orb, UAP, and unknown, if we could not extract the apparition type from the description. This would also allow us to see the frequency of each apparition sighting over the years and highlight if there were any years where apparitions were much more or less sighted than others. By using a stacked bar chart, we can clearly compare the differences across all years and see when certain apparition types are more or less frequent. The total apparition count is also more easily compared as the difference in height of the bars can easily be seen. This uses the work we did in the first assignment of parsing the apparition types from the haunted place descriptions. Through our visualization, we saw that most years had total apparition counts under around 60, and a few years (around 5) had much higher counts, with the maximum being around 145. Additionally, the most frequently seen apparition was ghosts across all years by a large margin, and the least sighted was the orb, although orb and UAP were both infrequent. Finally, there was a large portion of apparitions that were unknown, which was around half of the observations.
3. Number of Witnesses by Location
 - a. This visualization allowed us to use the witness counts we had extracted in the first assignment and use a map to see which locations across the country had the most and least witnesses across all the haunted place observations. This would allow us to see which locations likely had more haunted place observations in general, as we can assume

a higher witness count is associated with more reported hauntings, and infer which locations likely had more credible observations, as more witnesses may indicate a higher likelihood of a haunting actually occurring. Using a bubble chart allows us to easily visualize areas in the country that have much higher versus lower witness counts, and the usage of bubble size to correlate with witness count is an intuitive way for us to see that, too. Through this visualization, we saw that the northeast and midwest regions of the US had the highest number of witness counts, while the west had the least. In particular, the states Pennsylvania, Ohio, Illinois, Oklahoma, Arkansas, Minnesota, and New York had a few bubbles that were much larger than the other ones, indicating a much higher concentration of witnesses in specific cities in those states. This could either mean that some hauntings had a much higher witness count or that multiple hauntings occurred in the same location across the years. The high density of bubbles in states like Massachusetts, Connecticut, Rhode Island, and Michigan indicates many witnesses across the state through multiple different reports across the years.

4. Alcohol Abuse by State

- a. This visualization made use of a joined dataset documenting alcohol abuse across states. We specifically chose to use alcohol deaths per capita by state, as we thought this would give us a good idea of the severity of alcohol consumption across states and compare them. We believed a higher number would indicate less credibility in the hauntings reported in that state, as they could just be a result of impaired judgment from inebriated individuals. We used a color scale to indicate the deaths per capita, with a darker color meaning a higher number. Since we wanted to directly compare across states, we believe using a state choropleth would be most fitting, and a color scale would also be a straightforward way to compare different states. The resulting visualization showed that all numbers for the states were within a small range from 5.2 to 6.4, but it still gave us a better idea of the differences between states. Specifically, Connecticut and New Jersey had the least number of deaths, and West Virginia, followed by North Dakota, had the highest number. We see that states in the Midwest generally have a higher number of deaths, whereas states in the Northeast have lower numbers. Combined with our witness count visualization, this gives us further insight that the high number of witness counts in the Northeast may be more credible reporting due to lower alcohol abuse than compared to the high number of witness counts in the Midwest.

5. Top 10 States with the Most Hauntings Over Time

- a. We chose the D3 bar chart visualization to showcase the frequency of hauntings across states over time because it effectively illustrates the dynamic changes in haunting occurrences. The bar chart race format allows for an animated comparison of the most popular states with the highest frequencies of hauntings. We chose this because we wanted to challenge ourselves to build a story with our graph that showcased viewers a dynamic view of our entire dataset through a temporal lens. It also lets us identify trends regarding the reporting within each year. Assignment one played a huge role in forcing us to think about the importance of data extraction from our descriptions and Assignment 2 helped us extract entities in order to understand categorization. We also wanted to maybe see if specific decades/ time periods showcased more hauntings than others. We were hypothesizing there would be more hauntings in older years but realized there were significantly more during the more relevant decades. We assumed this may be due to the data addition and velocity being a lot more frequent in the current date than accumulating old records to add in.

II. New Visual Similarities Discovered Through Image Space

Q: Did Image Space allow you to find any similarity between the generated Haunted places images that previously was not easily discernible?

A:

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C:\USC Sem 2\USCI 550\4m2\space env\lib\site-packages\torchvision\models\_utils.py:288: UserWarning: The parameter "pretrained" is deprecated since 0.13 and may be removed in the future, please use "weights" instead.
d since 0.13 and may be removed in the future. The current behavior is equivalent to passing "weights=ResNet18_Weights.IMAGENET1K_V1". You can also use "weights=ResNet18_Weights.DEFAULT" to get the most up-to-date weights.
  warnings.warn(msg)

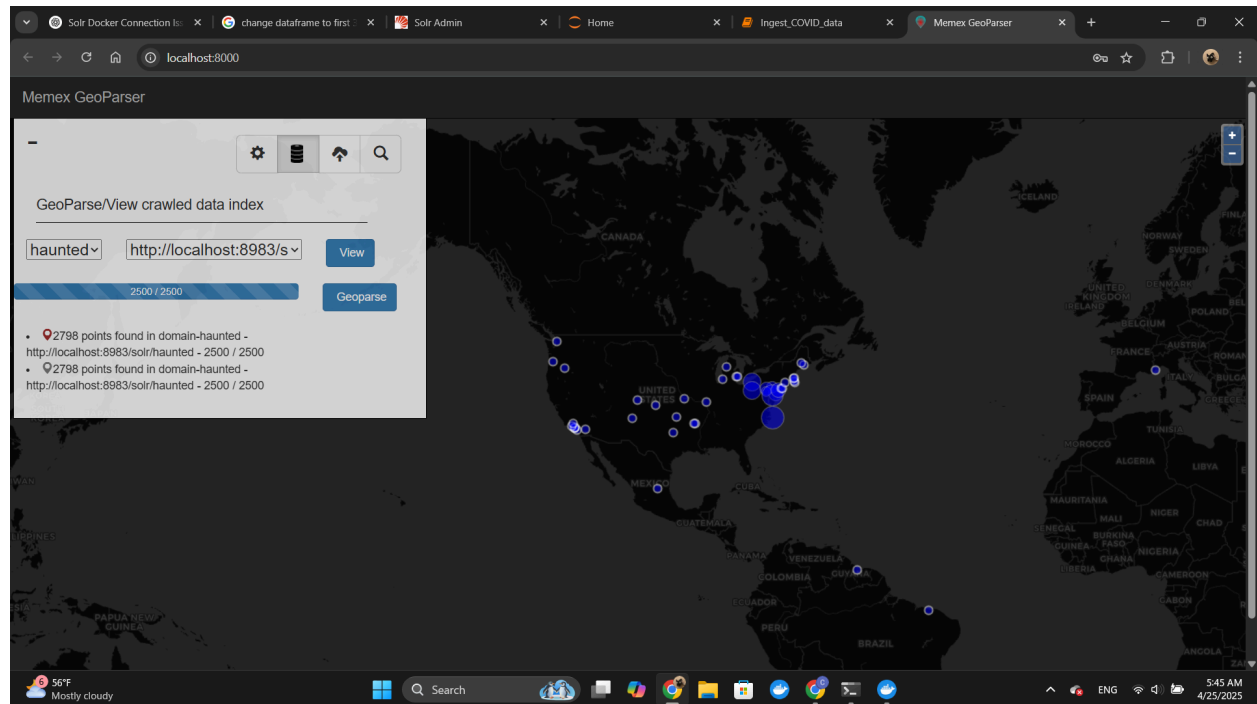
Top 5 similar images to images/woodland_park_colorado_10991.png:
1. woodland_park_colorado_10991 (distance=0.0000)
2. long_beach_california_2789 (distance=172.7488)
3. pikeville_tennessee_9376 (distance=173.7641)
4. comal_county_guene_texas_8891 (distance=177.4203)
5. pevely_missouri_9953 (distance=177.7458)
PS C:\USC Sem 2\USCI 550\4m2>
```

Image Space revealed similarities between haunted places that weren't immediately obvious through traditional viewing of the data from before. As shown in the screenshot, it identified the top five images most visually similar to "woodland_park_colorado_10991.png," including images from places like Long Beach, California, and Pikeville, Tennessee—locations that are geographically and culturally distinct and different. These connections, based on image embedding distance rather than geographic or textual features, suggest there may be shared visual or atmospheric traits across different haunted locations that wouldn't have been easily discovered through tabular data alone.

III. Analysis of Location Data and Emerging Correlations

Q: What type of location data showed up in your data? Any correlations not previously seen, e.g., from assignment 1?

A:



After using MEMEX GeoParser to explore the location data in our dataset (the “city”, “state”, “country”, “location”, “latitude”, and “longitude” columns), we were able to get a geographic visualization of the haunted places. The visualization showed that most haunted places were reported on the East Coast, specifically North Carolina, Delaware, and New Jersey. Having each haunted location plotted on a map made it much easier to identify patterns in the dataset than from previous assignments. Additionally, the GeoParser identified locations in other countries, although the dataset was supposed to only include locations in the US. This is likely due to the parser becoming confused with locations with similar names to other countries, such as French Valley, CA. We found similar results when using GeoTopicParser in Assignment 2.

IV. Reflections on Using Image Space and Image Cat

Q: Also include your thoughts about Image Space and ImageCat – what was easy about using them? What wasn't?

A:

Working with Image Space and ImageCat for this assignment was an interesting experience, both in terms of the potential these tools offer and the challenges that come with setting it up initially. When I first saw the assignment, I was excited about the idea of exploring haunted place images through forensic analysis and visual similarity. The Image Space platform, as described, sounded like a comprehensive tool, combining metadata extraction, image indexing, and even a similarity-based search UI backed by SMQTK.

The ImageCat component (which handles ingestion and metadata extraction) was fragmented across folders. I spent quite a bit of time trying to trace these scripts or figure out how to replicate the ImageSpace UI, only to realize that the best approach was to manually reconstruct the pipeline.

That's where things got interesting. I ended up using Tika-Python to extract image metadata and OCR content myself. Once I got past a few issues with malformed TSV entries and non-existent image paths, the ingestion process actually worked smoothly. Writing my own ETL pipeline turned out to be quite satisfying because I had full control over how the metadata was structured, cleaned, and saved. I stored it in a format that was compatible with Solr, and then manually indexed the data into a local Solr instance, which allowed me to browse and query the images through Solr's admin panel.

What was easy? Once Tika was set up, metadata extraction was simple and reliable. Writing my own ETL and Solr ingestion scripts gave me flexibility and control. Using SMQTK's descriptor generator (once dependencies were resolved) was a powerful way to explore image similarity. The local image locator was fast and efficient, especially for a technology developed almost a decade ago.

What was not easy? The initial setup and installation of requirements, as the PyTorch and Torchvision would not be compatible sometimes, causing issues. Luckily, I was using a GPU which made the process much faster.

One of the most satisfying parts of this project was building a working similarity search function based on the descriptors we generated with SMQTK. After extracting the image feature vectors using a ResNet18 model, I wrote a script that allowed me to input the path of a query image, and the application would then load its descriptor and compare it to all other descriptors in the dataset using cosine similarity. The results were surprisingly effective: haunted places with similar lighting, shapes, or textures often clustered together, even when the specific content was different. For example, foggy cemeteries or Victorian-style buildings tended to appear as top results when similar features existed in the query image. This part of the project gave me a clear, visual confirmation that the descriptor pipeline and indexing were working as intended.